

## RESEARCH PROFILE

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My research focuses on reliable decision making under uncertainty. I study how agents should act when the environment is stochastic, partially observable, adversarial, or not known exactly, using models such as stochastic games, Markov decision processes (MDPs), partially observable MDPs (POMDPs), and robust MDPs.

- **Mathematical foundations:** structure of optimal strategies, equilibria, robustness, and long-run objectives.
- **Algorithms and complexity:** efficient algorithms and tight complexity classifications for decision-making problems.
- **AI and learning:** learning and robust decision making under incomplete or misspecified models.

My research program aims to design agents with provable guarantees despite incomplete information, changing environments, and strategic interaction.

## EDUCATION

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- **Ph.D. in Computer Science** 2021 – Present  
Institute of Science and Technology Austria (ISTA), Klosterneuburg, Austria
- **B. Sc. in Computer Engineering** 2016 – 2021  
Sharif University of Technology, Tehran, Iran GPA: 18.93/20

## PUBLICATIONS

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- “Generalized Bidding Games: Where Bidding and Stochastic Games Meet”  
**Ali Asadi**, Thomas A. Henzinger, Ehsan Kafshdar Goharshady, Pavol Kebis, Kaushik Mallik  
in the *Proceedings of the 37<sup>th</sup> International Conference on Concurrency Theory (CONCUR’26)*, 2026.
- “PAC Learning in Turn-Based Stochastic Games with Reachability Objectives: A Decentralized Private Approach via Expected Conditional Distance”  
**Ali Asadi**, Krishnendu Chatterjee, Pavol Kebis  
in the *Proceedings of the 37<sup>th</sup> International Conference on Concurrency Theory (CONCUR’26)*, 2026.
- “Strongly Polynomial Time Complexity of Policy Iteration for  $L_\infty$  Robust MDPs”  
**Ali Asadi**, Krishnendu Chatterjee, Ehsan Goharshady, Mehrdad Karrabi, Alipasha Montaseri, Carlo Pagano  
in the *Proceedings of the 39<sup>th</sup> Annual Conference on Learning Theory (COLT’26)*, PMLR 336:427–457, 2026.
- “Qualitative Analysis of  $\omega$ -Regular Objectives on Robust MDPs”  
**Ali Asadi**, Krishnendu Chatterjee, Ehsan Goharshady, Mehrdad Karrabi, Ali Shafiee  
in *The 40<sup>th</sup> Annual AAAI Conference on Artificial Intelligence (AAAI’26)*.
- “Revealing POMDPs: Qualitative and Quantitative Analysis for Parity Objectives”  
**Ali Asadi**, Krishnendu Chatterjee, David Lurie, Raimundo Saona  
in *The 40<sup>th</sup> Annual AAAI Conference on Artificial Intelligence (AAAI’26)*.
- “ $\varepsilon$ -Stationary Nash Equilibria in Multi-player Stochastic Graph Games”  
**Ali Asadi**, Léonard Brice, Krishnendu Chatterjee, K. S. Thejaswini  
in the *45<sup>th</sup> IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS’25)*.
- “Limit-sure reachability for small memory policies in POMDPs is NP-complete”  
**Ali Asadi**, Krishnendu Chatterjee, Raimundo Saona, Ali Shafiee  
in the *41<sup>st</sup> Conference on Uncertainty in Artificial Intelligence (UAI’25)*.
- “Lower Bound on Howard Policy Iteration for Deterministic Markov Decision Processes”  
**Ali Asadi**, Krishnendu Chatterjee, Jakob de Raaij  
in the *41<sup>st</sup> Conference on Uncertainty in Artificial Intelligence (UAI’25)*.

- “Concurrent Stochastic Games with Stateful-discounted and Parity Objectives: Complexity and Algorithms”  
**Ali Asadi**, Krishnendu Chatterjee, Raimundo Saona, Jakub Svoboda  
in the 44<sup>th</sup> IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (**FSTTCS’24**).
- “Deterministic Sub-exponential Algorithm for Discounted-sum Games with Unary Weights”  
**Ali Asadi**, Krishnendu Chatterjee, Raimundo Saona, Jakub Svoboda  
in the Proceedings of the 39<sup>th</sup> Annual Symposium on Logic in Computer Science (**LICS’24**), 2024.
- “Polynomial reachability witnesses via Stellensätze”  
**Ali Asadi**, Krishnendu Chatterjee, Hongfei Fu, Amir Kafshdar Goharshady, Mohammad Mahdavi  
in the Proceedings of the 42<sup>nd</sup> ACM SIGPLAN International Conference on Programming Language Design and Implementation (**PLDI’21**), 2021.
- “Faster Algorithms for Quantitative Analysis of MCs and MDPs with Small Treewidth”  
**Ali Asadi**, Krishnendu Chatterjee, Amir Kafshdar Goharshady, Kiarash Mohammadi, Andreas Pavlogiannis  
in the Proceedings of the 18<sup>th</sup> International Symposium on Automated Technology for Verification and Analysis (**ATVA’20**), 2020.

## MANUSCRIPTS UNDER REVIEW

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- “The Complexity of Approximating the Value in Revealing POMDPs with Long-Run Average Objectives”  
**Ali Asadi**, Krishnendu Chatterjee, David Lurie  
submitted to *SIAM Journal on Control and Optimization* (**SICON**).
- “Algorithms with Smoothed Polynomial-Time Complexity for Deterministic Discounted-sum and Mean-payoff Games”  
**Ali Asadi**, Krishnendu Chatterjee, Ruichen Luo  
submitted to the 38<sup>th</sup> Annual ACM-SIAM Symposium on Discrete Algorithms (**SODA’27**).
- “Values of Absorbing Recursive Games Express All Real Algebraic Numbers”  
**Ali Asadi**, Krishnendu Chatterjee  
submitted to *Mathematics of Operations Research* (**MOOR**).
- “On the Complexity of Discounted Robust MDPs with  $L_p$  Uncertainty Sets”  
**Ali Asadi**, Krishnendu Chatterjee, Alipasha Montaseri, Ali Shafiee  
submitted to the 40<sup>th</sup> Annual Conference on Neural Information Processing Systems (**NeurIPS’26**).

## TALKS AND POSTERS

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- “Revealing POMDPs: Qualitative and Quantitative Analysis for Parity Objectives”  
**Talk:** Highlights of Logic, Games and Automata (Highlights’26), Vienna, Austria Sept. 2026  
**Poster:** Paris Game Theory Festival 2026, Paris, France June 2026  
**Poster:** AAAI Conference on Artificial Intelligence (AAAI’26), Singapore Jan. 2026  
**Seminar talk:** Combinatorics, Games and Optimisation Seminar, LSE, London, UK Dec. 2025
- “Limit-sure reachability for small memory policies in POMDPs is NP-complete”  
**Poster:** 30 Years of Game Theory at Institut Henri Poincaré, Paris, France Oct. 2025
- “Deterministic Sub-exponential Algorithm for Discounted-sum Games with Unary Weights”  
**Talk:** International Conference on Reachability Problems (RP’24), Vienna, Austria Sept. 2024  
**Talk:** ACM/IEEE Symposium on Logic in Computer Science (LICS’24), Tallinn, Estonia July 2024  
**Poster:** Mediterranean Game Theory Symposium 2024, Marseille, France May 2024

## HONORS AND AWARDS

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- **Silver medal** in the 10<sup>th</sup> Asia-Pacific Informatics Olympiad (APIO) 2016
- **Gold and Silver medals** in the 25<sup>th</sup> and 24<sup>th</sup> Iranian National Olympiad in Informatics (INOI) 2014, 2015
- **2<sup>nd</sup> place** in the ACM-ICPC West Asia Regional Contest, Tehran, Iran. 2016
- Iranian National Elites Foundation grant for undergraduate studies 2014 – 2021
- Research Scholarship from the Austrian Exchange Agency (OeAD-GmbH) 2019

## SCIENTIFIC SERVICE

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- **Program Committee Member**
  - AAAI Conference on Artificial Intelligence (AAAI'27) 2027
  - AAAI Conference on Artificial Intelligence (AAAI'26) 2026
- **Journal Reviewer**
  - SIAM Journal on Control and Optimization (SICON) 2026
  - Information and Computation 2026
- **Conference Reviewer**
  - Conference on Uncertainty in Artificial Intelligence (UAI'26) 2026
  - Foundations of Software Technology and Theoretical Computer Science (FSTTCS'25) 2025
  - International Symposium on Mathematical Foundations of Computer Science (MFCS'25) 2025
  - International Colloquium on Automata, Languages, and Programming (ICALP'24) 2024

## TEACHING EXPERIENCE

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### University Teaching

- **Teaching Assistant, Institute of Science and Technology Austria (ISTA)** 2024 – 2025  
Held office hours and prepared and graded homework assignments.
  - Computer Science Track Core Course Spring 2024; Spring 2025
  - Formal Methods: Algorithmic Approaches Spring 2024
- **Teaching Assistant, Sharif University of Technology** 2017 – 2018  
Prepared and graded homework assignments.
  - Data Structures and Algorithms Spring 2018
  - Probability and Statistics Fall 2017

### Olympiad Teaching and Educational Service

- **Lecturer, Algorithms and Data Structures** 2016 – 2019  
Prepared students for the Iranian National Olympiad in Informatics (INOI).
  - Allameh Helli High School 2016 – 2018
  - Allameh Tabatabaee High School 2016 – 2019
- **Scientific Committee Member and Problem Author, INOI**
  - INOI Final Exams 2022 – 2025
  - IOI Team Selection Exams 2017
  - INOI Summer Camp Programming Exams 2016

## RESEARCH SUPERVISION

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### Research Interns and Rotation Students, Institute of Science and Technology Austria (ISTA)

- **Pavol Kebis** Fall 2025
  - “PAC Learning in Turn-Based Stochastic Games with Reachability Objectives: A Decentralized Private Approach via Expected Conditional Distance” (**CONCUR’26**)
- **Jack Quirion** Summer 2025
  - “Multi-Objective Reachability in Stochastic Games” (**planned submission to ICALP’27**)
- **Alipasha Montaseri** Summer 2025
  - “Strongly Polynomial Time Complexity of Policy Iteration for  $L_\infty$  Robust MDPs” (**COLT’26**)
  - “On the Complexity of Discounted Robust MDPs with  $L_p$  Uncertainty Sets” (**submitted to NeurIPS’26**)
- **Ali Shafiee** Summer 2024
  - “Qualitative Analysis of  $\omega$ -Regular Objectives on Robust MDPs” (**AAAI’26**)
  - “Limit-sure reachability for small memory policies in POMDPs is NP-complete” (**UAI’25**)
- **Jakob de Raaij** Summer 2023
  - “Lower Bound on Howard Policy Iteration for Deterministic Markov Decision Processes” (**UAI’25**)

## INDUSTRY EXPERIENCE

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- **Engineering Manager, Divar Co.** Sep. 2019 – Sep. 2021  
Led a team of 10 engineers building scalable advertising infrastructure.
- **Software Engineer, CafeBazaar Co.** July 2017 – July 2019  
Developed high-scale payment and deployment systems.